Fundamentals 14

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1a. Let $f(x) = 0$. We claim that this converges to 0. Fix $\varepsilon > 0$. Then for all $x \in \mathbb{R}$, $|f(x) - 0| = 0 < \varepsilon$.

Now, let $f(x) = x$. We claim that this does not converge. To prove this, we must show that for all L, there exists $\varepsilon > 0$ such that for all M there exists $x > M$ such that $|x - L| \geq \varepsilon$. Fix L, and let $\varepsilon = 1$. Then choose $x = \max(L + 1, M + 1)$. We have that $|x - L| \geq |L + 1 - L| = 1 \geq \varepsilon$.

1b. Define f as

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Z} \\ x & \text{otherwise.} \end{cases}$$

Since $f(n) = 0$ for all integers, $f(n) \to 0$, but $f(x)$ does not converge.

1c. We say that $f(x) \to \infty$ as $x \to x_0$ if for all N, there exists $\delta > 0$ such that if $|x - x_0| < \delta$ then $f(x) > N$.

We say that $f(x) \to \infty$ as $x \to \infty$ if for all N, there exists M such that if $x > M$ then $f(x) > N$.

2. Define f as

$$f(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n} \text{ for some } n \in \mathbb{N} \\ 0 & \text{otherwise.} \end{cases}$$

We have that $f(1/n) = 1$ whenever $n \in \mathbb{N}$, so $\lim_{n\to\infty} f(1/n) = 1$. However, $\lim_{x\to 0} f(x)$ doesn't exist, since it is 0 at all irrational points near 0 and it is 1 at many rational points near 0.

3a. Fix ε. For all x, y, $|f(x) - f(y)| = |c - c| = 0 < \varepsilon$.

3b. Fix ε and let $\delta = \varepsilon$. If $|x - y| < \delta$ then $|f(x) - f(y)| = |x - y| < \delta = \varepsilon$.

3c. We will prove this using induction on the degree of the polynomial. First, we have that $f(x) = c$ is continuous. Now, assume that all polynomials

of degree $\leq n$ are continuous. We wish to show that all polynomials of degree $n+1$, given by

$$f(x) = \sum_{k=0}^{n+1} a_k x^k,$$

are continuous. We know that the polynomials

$$p(x) = \sum_{k=0}^{n} a_{k+1} x^k, \text{ and } q(x) = a_0$$

are continuous by the induction hypothesis. This means that

$$\lim_{x \to y} p(x) = p(y), \text{ and } \lim_{x \to y} q(x) = q(y).$$

Since $f(x) = xp(x) + q(x)$, we have that

$$\lim_{x \to y} f(x) = \lim_{x \to y} (xp(x) + q(x)) = \left(\lim_{x \to y} x\right)\left(\lim_{x \to y} p(x)\right) + \left(\lim_{x \to y} q(x)\right)$$

$$= yp(y) + q(y) = f(y).$$

Therefore, f is continuous.

3d. Yes, $f(x) = \frac{1}{x}$ is continuous because it is continuous at all points in its domain, $\mathbb{R}\backslash\{0\}$.

4. We wish to show that there exists ε such that for all $\delta > 0$ there exists $|x| < \delta$ such that $\left|\frac{1}{x} - r\right| \geq \varepsilon$. Fix $\varepsilon > 0$, fix $\delta > 0$, and choose x such that $|x| \leq \frac{1}{r+\varepsilon}$. Then

$$\left|\frac{1}{x} - r\right| \geq \left|\frac{1}{1/(r+\varepsilon)} - r\right| = |r + \varepsilon - r| = \varepsilon.$$

5a. Fix $\varepsilon > 0$. There exists $\delta > 0$ such that $|x - x_0| < \delta \implies |f(x) - L| < \varepsilon/c$. So if $|x - x_0| < \delta$, then

$$|cf(x) - cL| = c|f(x) - L| < c \cdot \varepsilon/c = \varepsilon.$$

5b. Fix $\varepsilon > 0$. There exist $\delta_1, \delta_2 > 0$ such that $|x - x_0| < \delta_1 \implies |f(x) - L| < \varepsilon/2$ and $|x - x_0| < \delta_2 \implies |g(x) - D| < \varepsilon/2$. Let $\delta = \min(\delta_1, \delta_2)$. Then if $|x - x_0| < \delta$,

$$|f(x) + g(x) - (L + D)| \leq |f(x) - L| + |g(x) - D| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

5c. Fix $\varepsilon > 0$. Since f converges as $x \to x_0$, $|f(x)|$ is bounded in some region R around x_0. So let $M > \max|f(x)|$ and let $N > D$. There exist $\delta_1, \delta_2 > 0$ such that $|x - x_0| < \delta_1 \implies |f(x) - L| < \varepsilon/2M$ and $|x - x_0| < \delta_2 \implies |g(x) - D| <$

$\varepsilon/2N$. We can update δ_1 so that $(x_0-\delta_1, x_0+\delta_1) \subseteq R$. Now, let $\delta = \min(\delta_1, \delta_2)$. If $|x - x_0| < \delta$,

$$|f(x)g(x)-LD| = |f(x)g(x)-f(x)D+f(x)D-LD| = |f(x)(g(x)-D)-D(f(x)-L)|$$

$$\leq |f(x)||g(x) - D| - |D||f(x) - L| < M\frac{e}{2M} + N\frac{\varepsilon}{2N} = \varepsilon.$$

5e. Let $M > \max|g(x)|$ and fix $\varepsilon > 0$. We know that there exists $\delta > 0$ such that if $|x-x_0| < \delta$ then $|f(x)| < \varepsilon/M$. Then $|f(x)g(x)| < M|f(x)| < \varepsilon/M$.

6. We know that $\lim_{x\to x_0} f(x) = f(x_0)$, so we wish to show that $f(x_0) = L$. For contradiction, assume $L \neq f(x_0)$, and let $\varepsilon < |L - f(x_0)|$.

Since f is continuous at x_0, there exists $\delta_1 > 0$ such that if $|x - x_0| < \delta_1$ then $|f(x) - f(x_0)| < \varepsilon$. Also, we are given that there exists $\delta_2 > 0$ such that if x is rational and $|x - x_0| < \delta_2$ then $|f(x) - L| < \varepsilon$. Let x be rational and $< \min(\delta_1, \delta_2)$. Then

$$|f(x) - f(x_0)| < \varepsilon, \quad |f(x) - L| < \varepsilon.$$

We also have that

$$|L - f(x_0)| = |f(x) - f(x_0) - (f(x) - L)| \leq |f(x) - f(x_0)| + |f(x) - L| < 2\varepsilon.$$

But $\varepsilon < |L - f(x_0)|$, so this is a contradiction. Therefore, $f(x_0) = L$.

7. Assume that for all continuous functions g, $g \circ f$ is continuous at x. Then if $g(x) = x$, $g \circ f = f$ is also continuous at x.

Now, assume that f is continuous at x. Let $g : \mathbb{R} \to \mathbb{R}$ be any continuous function. Since g is continuous at $f(x)$, by Proposition 14.13, $g \circ f$ is continuous at x.

8. I think it makes sense to let all such functions be continuous, since the idea behind continuity, that small changes in x result in small changes of $f(x)$, is trivially satisfied when the domain consists of isolated points. Discontinuities should generally occur when the limit to a point x doesn't match $f(x)$. But there are no limit points since the domain is discrete, and since points in the domain can't be arbitrarily close to other points.

9. Define $g, h : (-\infty, 0) \to \mathbb{R}$ as

$$g(x) = \begin{cases} x + 5 & \text{if } x \leq -5 \\ x^2 - 25 & \text{if } -5 < x < 0, \end{cases}$$

$$h(x) = \begin{cases} x^2 - 25 & \text{if } x \leq -5 \\ x + 5 & \text{if } -5 < x < 0. \end{cases}$$

If $x \leq -5$, we have that $x + 5 \leq x + 5$ and $x + 5 \geq x^2 - 25$, so $g(x) \leq f(x)$. If $x > 5$, then $x^2 - 25 \leq x^2 - 25$ and $x^2 - 25 < x + 5$, so $g(x) \leq f(x)$. Therefore, $g(x) \leq f(x)$ for all $x < 0$. Similarly, $h(x) \geq f(x)$.

Since g and h are continuous ($x + 5$ and $x^2 - 25$ are continuous and they have the same limit at $x = -5$),

$$\lim_{x \to -5} g(x) = g(-5) = 0,$$

$$\lim_{x \to -5} h(x) = h(-5) = 0,$$

so $\lim_{x \to -5} f(x) = 0$.

10. We need to show that for all L, there exists ε such that for all $\delta > 0$ there exists x with $|x| < \delta$ such that $|\sin(1/x) - L| \geq \varepsilon$. Fix L such that $|L| \leq 1$ (otherwise the result is trivial, since L is always a finite distance away from all values of $\sin(1/x)$), let $\varepsilon = 1$, and fix $\delta > 0$.

Let $y = L - 1$ if $L > 0$ or $L + 1$ if $L \leq 0$. Now, find $z > 0$ such that $\sin(z) = y$ and $z > \frac{1}{\delta}$. We know this exists, since sin periodically reaches all points in $[-1, 1]$ infinitely many times. Then let $x = \frac{1}{z}$ so that $z = \frac{1}{x} > \frac{1}{\delta}$, and $|x| < \delta$. Then

$$\left| \sin\left(\frac{1}{x}\right) - L \right| = |\sin(z) - L| = |y - L| \geq 1 = \varepsilon,$$

as desired.

11a. We prove this by induction. When $n = 0$ the inequality holds, as $(1 + x)^0 = 1 \geq 1 + 0x = 1$.

Now, assume that $(1 + x)^n \geq 1 + nx$. Then

$$(1 + x)^{n+1} \geq (1 + nx)(1 + x) = 1 + n(x + 1) + nx^2,$$

So $(1 + x)^{n+1} \geq 1 + n(x + 1)$.

11b. We claim that $\frac{1}{n!} \leq \frac{1}{2^{n-1}}$ for all $n \geq 0$ and prove this by induction. We know that $\frac{1}{0!} \leq \frac{1}{2^{-1}} = 2$, so the base case holds. Now, assume that $\frac{1}{n!} \leq \frac{1}{2^{n-1}}$. Then, since $n \geq 1$,

$$\frac{1}{(n+1)!} \leq \frac{1}{2n!} \leq \frac{1}{2 \cdot 2^{n-1}} = \frac{1}{2^n},$$

as desired.

Therefore, we have that

$$\sum_{j=0}^{n} \frac{1}{j!} \leq \sum_{j=0}^{n} \frac{1}{2^{j-1}} = \frac{2}{1 - \frac{1}{2}} = 4,$$

so the sum is bounded above.

11c. We have that

$$\left(1+\frac{1}{n}\right)^n = \sum_{j=0}^{n} \binom{n}{j} \left(\frac{1}{n}\right)^j.$$

So it suffices to show that

$$\binom{n}{j} \left(\frac{1}{n}\right)^j \leq \frac{1}{j!}$$

for all $j \geq 0$. Expanding the LHS and working backwards, we get

$$\binom{n}{j} \left(\frac{1}{n}\right)^j = \frac{n!}{j!(n-j)!} \cdot \frac{1}{n^j} \leq \frac{1}{j!},$$

$$\frac{n!}{(n-j)!} \leq n^j,$$

$$n(n-1)\cdots(n-j+1) \leq n^j,$$

which is true, since the LHS multiplies j terms that are each $\leq n$ and the RHS multiplies n j times. So x_n is bounded.